

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Digital Signal and Image Processing (01CT1513)	Aim : Open Ended Problem 1	
Open Ended Problem : 1	Date: 15/07/2026	Enrolment No: 92400133125

GitHub Link : <https://github.com/mankumarbhalodiya/Digital-Signal-and-Image-Processing/tree/main/Experiment%201>

Problem 1 :

```
import numpy as np

import matplotlib.pyplot as plt

n = np.arange(-3, 7)

x = [2,3,2,3,2,3,2,3,2,3]

plt.stem(n, x)

plt.title("Signal 1")

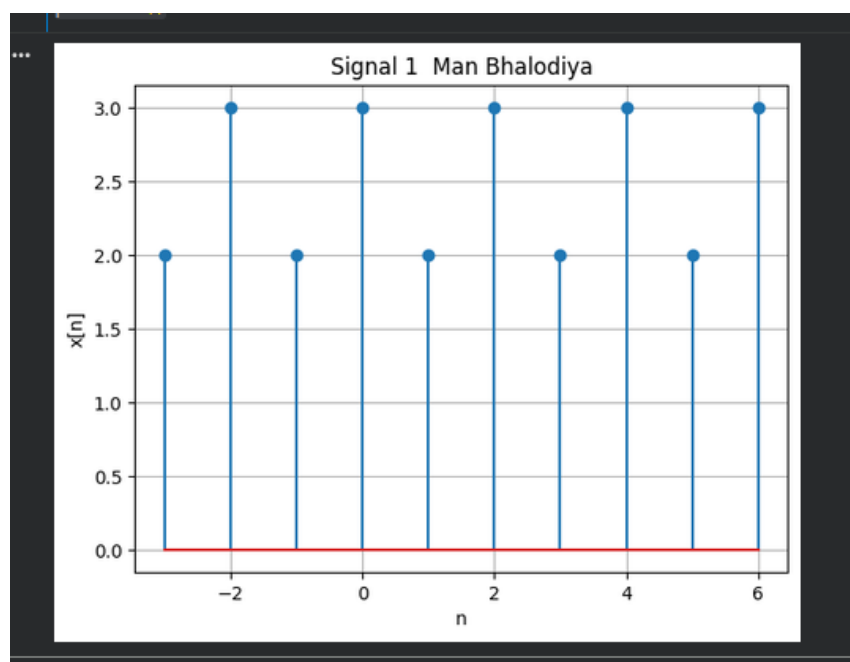
plt.xlabel("n")

plt.ylabel("x[n]")

plt.grid(True)

plt.show()
```

Output :



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Problem 2 :

```
import numpy as np

import matplotlib.pyplot as plt

n = np.arange(0, 7)

x = [1,2,3,3,2,1,0]

plt.step(n, x, where='post')

plt.xlim(0,6)

plt.ylim(0,4)

plt.title("Signal 2")

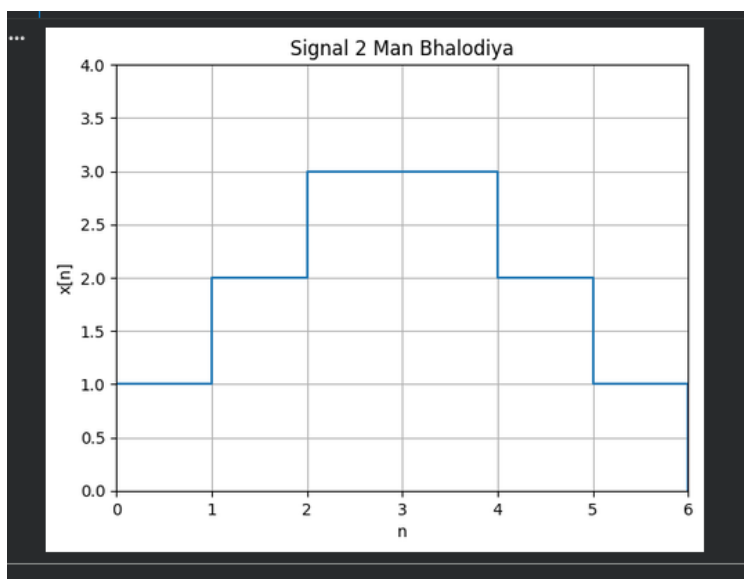
plt.xlabel("n")

plt.ylabel("x[n]")

plt.grid(True)

plt.show()
```

Output :



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Problem 3 :

```

import numpy as np

import matplotlib.pyplot as plt

x1 = np.linspace(-2,-1,100)

y1 = -2*np.ones(100)

x2 = np.linspace(-1,1,100)

y2 = 2*x2

x3 = np.linspace(1,2,100)

y3 = 2*np.ones(100)

plt.plot([-2,-2],[0,-2])    # Vertical line at x = -2

plt.plot(x1,y1)

plt.plot(x2,y2)

plt.plot(x3,y3)

plt.plot([2,2],[2,0])    # Vertical line at x = 2

plt.axhline(0,color='black')

plt.axvline(0,color='black')

plt.xlim(-2.5,2.5)

plt.ylim(-3,3)

plt.grid(True)

plt.title("Signal 3")

plt.xlabel("n")

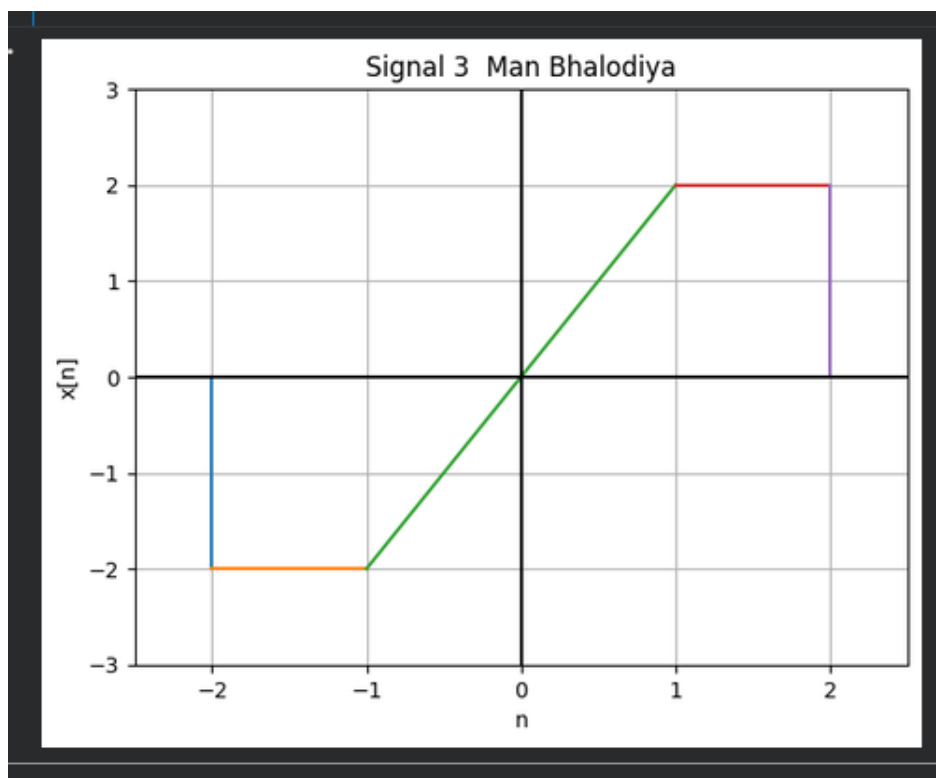
plt.ylabel("x[n]")

plt.show()

```

Output :

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Problem 4 :

```

import numpy as np
import matplotlib.pyplot as plt

n = np.arange(-2, 12)

u = lambda x: np.where(x >= 0, 1, 0)

x = u(n) - u(n-3) - 5*u(n-7)

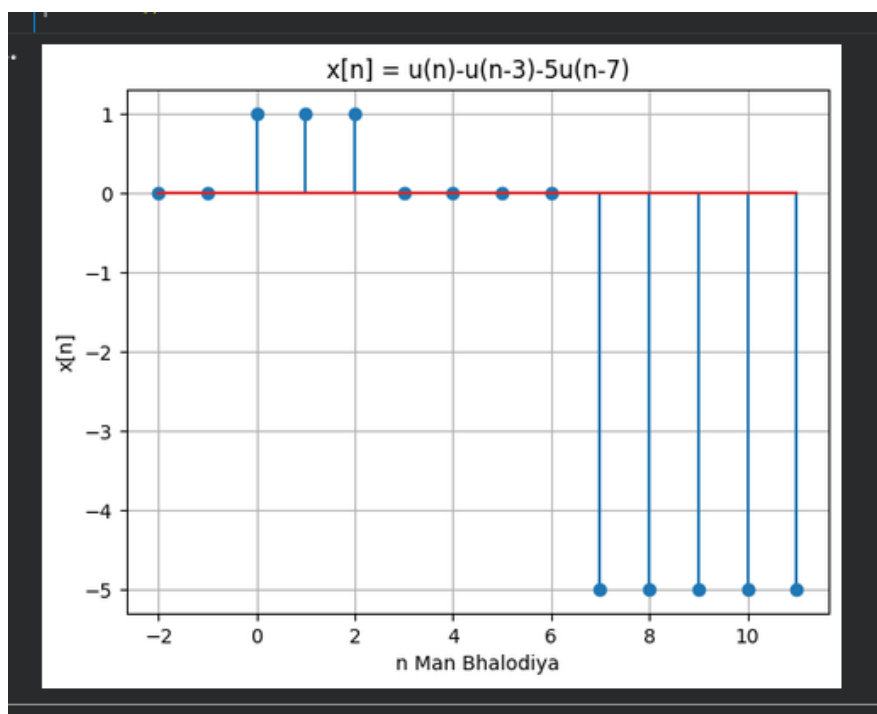
plt.stem(n, x)

plt.title("x[n] = u(n)-u(n-3)-5u(n-7)")
plt.xlabel("n Man Bhalodiya")
plt.ylabel("x[n]")
plt.grid(True)
plt.show()

```

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Output :



Problem 5 :

```
import numpy as np

import matplotlib.pyplot as plt

n = np.arange(-3, 4)

delta = lambda x:
    np.where(x == 0, 1, 0)

x = delta(n) + 3delta(n-1) + 5delta(n+1)

plt.stem(n, x)

plt.title("x[n]=δ(n)+3δ(n-1)+5δ(n+1)")

plt.xlabel("n Man Bhalodiya")
```

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```
plt.ylabel("x[n]")
```

```
plt.grid(True)
```

```
plt.show()
```

Output :

